

TAPAN KHADELWAL

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Profile Summary

AI/ML Engineer, AWS Certified (AI Practitioner, Cloud Practitioner), building and deploying GenAI and Agentic AI systems end to end. Built multi-agent pipelines with LangChain and LangGraph, including a research assistant and a 5-agent trip planner deployed on AWS, plus RAG systems using vector databases (Chroma) and embedding models with prompt-injection defense (5/5 adversarial payloads held). Hands-on with AWS (EC2, S3, Lambda, DynamoDB, Cognito, SageMaker, Bedrock), SQL/NoSQL databases (PostgreSQL, MySQL, MongoDB), Python, NLP (Hugging Face Transformers, NLTK), scikit-learn, Docker, and Git.

Education

Jagannath University

B.Tech in Computer Science (AI/ML) – CGPA: 8.0

Jaipur, India

Sep 2022 – June 2026

AI/ML Projects

Multi-Agent Research System

Tech: Python, LangChain, LangGraph, Mistral AI (LLM), Streamlit

multi-agent-research-system-vzra.onrender.com

- Designed and built a modular, multi-stage Agentic AI pipeline (Search, Reader, Writer, Critic) using LangChain and LangGraph, with tool-calling agents and LLM API integration (Mistral AI) to autonomously research topics and generate structured reports.
- Engineered prompts and agent-to-agent workflows to coordinate retrieval, synthesis, and critique steps, deploying the full pipeline as a live, user-facing Streamlit web app.

Smart ChatBot for Student Query Assistance

Tech: Python, Flask, NLP, Hugging Face Transformers, NLTK

ai-based-student-chatbot.onrender.com

- Built and deployed a Flask-based backend serving a conversational AI assistant, handling real-time student queries across 8 categories (exams, fees, hostel, etc.) using NLP-based, confidence-scored response matching.
- Applied Hugging Face Transformers and NLTK for text preprocessing and intent handling, packaging the assistant as a live web application.

Professional Experience

AI/ML Engineer

June 2025 – May 2026

Cognus Technology

Jaipur

- Built a Python NLP text-classification pipeline (scikit-learn, TF-IDF) on ~3,000 labeled support tickets across 6 categories, lifting weighted F1 from a 0.71 keyword-matching baseline to 0.86.
- Automated triage of incoming requests by auto-routing high-confidence predictions and escalating low-confidence ones for human review, cutting manual triage volume by an estimated 25–30%.
- Iterated on preprocessing (tokenization, n-gram/TF-IDF tuning) and model selection with the team, then moved the classifier from a notebook prototype to a deployable script using Git for version control.

AI/ML Engineer Intern

July 2024 – Aug 2024

iGeeks Technology

Bangalore

- Cleaned and prepared a ~5,000-row real-world dataset in Python/SQL – handling missing values, outliers, and categorical encoding ahead of model training.
- Ran feature engineering and 5-fold cross-validation across three model types (Logistic Regression, Random Forest, Gradient Boosting), improving validation accuracy from a 71% baseline to 89% on the held-out set.
- Built automated logging around the deployed model's predictions in production (~2,000 daily inference requests) to flag data-quality issues and prediction drift for the team.

Skills

GenAI & Agentic AI: LangChain, LangGraph, Multi-Agent Pipelines, Tool-Calling Agents, MCP (Model Context Protocol), LLM API Integration (Mistral AI, Groq), Ollama (Local LLMs), Prompt Design, LLM Guardrails

RAG & Vector Search: Retrieval-Augmented Generation (RAG), Vector Databases (Chroma), Sentence-Transformers Embeddings, RAG Evaluation (Faithfulness, Citation Accuracy)

Deep Learning & Computer Vision: PyTorch, torchvision, CNNs (EfficientNet), OpenCV, Transfer Learning, Model Training & Fine-Tuning

NLP: Hugging Face Transformers, NLTK, Text Classification, Intent/Response Matching

Backend & Automation: FastAPI, Flask, REST APIs, Webhooks, JWT/JWKS Auth, boto3, n8n (Workflow Automation)

Programming: Python, Java, C++

Data & Databases: SQL, MySQL, PostgreSQL, MongoDB, DynamoDB, Query Optimization, Data Cleaning, EDA

ML & Model Evaluation: Scikit-learn, Cross-Validation, Statistical Model Evaluation (Precision, Recall, F1-score)

Cloud & DevOps: AWS (EC2, S3, Lambda, API Gateway, IAM, DynamoDB, Cognito, SageMaker, Bedrock), Docker, Git, GitHub

BI & Visualization: Power BI, Tableau, Dashboard Creation, Data Visualization, Business Reporting

Certifications

- AWS Certified AI Practitioner** – Amazon Web Services, issued Sep 2026 (ID: 13e2f2002bff4c63b023c47935823ae9).
- AWS Certified Cloud Practitioner** – Amazon Web Services, issued Sep 2026 (ID: b876ec05c391414b8586b4f52c03837b).
- Networking Essentials** – Cisco Networking Academy – IPv4/IPv6, DHCP, LAN setup, network security.